

An alternative to the H-coefficient technique for post-quantum security proofs

Christian Majenz

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Based on a line of work with Gorjan Alagic, Chen Bai, Jonathan Katz, Patrick Struck and Kayian Shi, eprint 2021/1601, 2022/1097, 2026/209

Motivation

When is post-quantum security for symmetric-key cryptography different?

Example: Block Ciphers

- Block cipher

A block cipher $E: \{0,1\}^m \times \{0,1\}^n \rightarrow \{0,1\}^n$ is a pseudorandom permutation. In particular, $E_k(\cdot) = E(k, \cdot)$ is a permutation of $\{0,1\}^n$ for all $k \in \{0,1\}^m$.

- n is the block size and m is the key size
- Example: AES

Motivation

AES is widely assumed to be post-quantum secure...

...so what is there even to study?

Security proofs for higher level constructions! Idealized models need revisiting.

⇒ Analysis needs to be done in Quantum Ideal Permutation/Cipher model!!!

Attack Models

What is the right definition of post-quantum security?

(Quantum) Ideal Permutation Model (QIPM)

- Model:
 - Public permutation P is replaced by **uniformly random permutation**.
 - All parties have **oracle access** to E and E^{-1}
 - Quantum: Need to provide **quantum oracle**

$$|k\rangle|x\rangle|y\rangle \mapsto |k\rangle|x\rangle|y \oplus E_k(x)\rangle \quad |k\rangle|x\rangle|y\rangle \mapsto |k\rangle|x\rangle|y \oplus E_k^{-1}(x)\rangle$$

Q1 vs Q2 Query Access (within QICM)

- QICM: quantum ideal cipher oracle.
- Adversary is granted oracle access to the keyed construction.
- Q1 vs Q2: different access to the keyed construction.

	Q1 (Post-quantum security)	Q2 (overkill for most things)
Adversary type	Quantum	Quantum
Queries to the ideal permutation P	Quantum (QIPM)	Quantum (QIPM)
Queries to keyed construction E_k	Classical	Quantum

The H-coefficient technique

The Swiss army knife of provable symmetric-key cryptography

The H-coefficient technique

- Adversary $\mathcal{A}$ interacting with one of two oracles, $\mathcal{O}_0$ or $\mathcal{O}_1$ drawn from probability distribution D_0 or D_1 , needs to guess which
- Advantage: $|\Pr[1 \leftarrow \mathcal{A}^{\mathcal{O}_0}] - \Pr[1 \leftarrow \mathcal{A}^{\mathcal{O}_1}]|$
- $\mathcal{O}_i$ can be multiple oracles
- Covers many situations, e.g. cipher constructions from ideal permutation/cipher

The H-coefficient technique

- Adversary $\mathcal{A}$ interacting with one of two oracles, $\mathcal{O}_0$ or $\mathcal{O}_1$ drawn from probability distribution D_0 or D_1 , needs to guess which
- $\rightsquigarrow$ Query transcript $T \in \mathcal{T}$

Observations:

- WLOG deterministic $\mathcal{A}$
- T determines the output bit
- RVs T_0, T_1
- $\Pr[T_i = \tau] \in \{0, p_i(\tau)\}$ where p_i only depends on $\mathcal{O}_i$

The H-coefficient technique

Theorem:

Let $B \subset \mathcal{T}$. Then

$$|\Pr[1 \leftarrow \mathcal{A}^{\mathcal{O}_0}] - \Pr[1 \leftarrow \mathcal{A}^{\mathcal{O}_1}]| \leq \Pr[T_1 \in B] + \varepsilon,$$

- Where ε is such that for all $\tau \notin B$ with $\Pr[T_0 = \tau] > 0$,

$$\frac{\Pr[T_0 = \tau]}{\Pr[T_{1\cdot} = \tau]} \geq 1 - \varepsilon$$

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More observations:

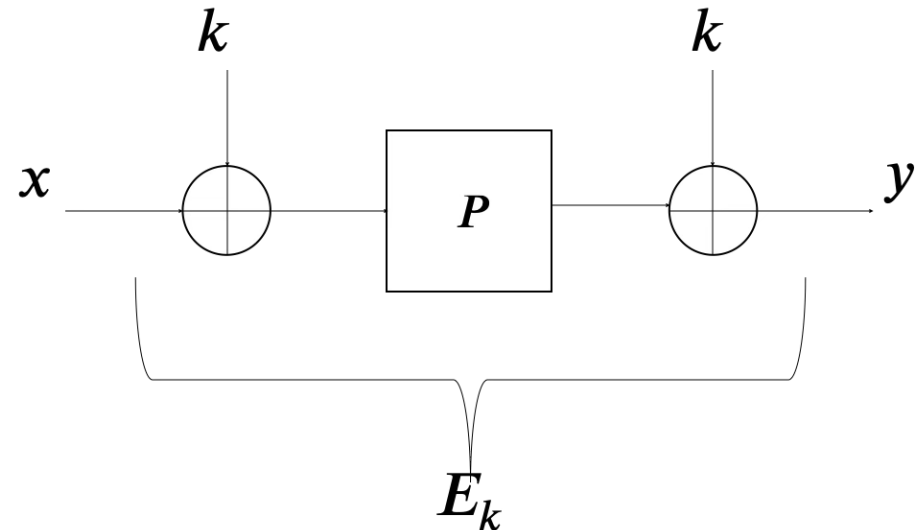
- Say $\mathcal{O}_0 / \mathcal{O}_1$ is real/ideal world: only need $\Pr[T_1 \in B]$ in ideal world
- $\Pr[T_i = \tau] \in \{0, p_i(\tau)\}$ where p_i only depends on $\mathcal{O}_i \rightsquigarrow$ no need to reason about adversary!

Example: Even-Mansour

Given a public permutation $P: \{0,1\}^n \rightarrow \{0,1\}^n$ and a key $k \in \{0,1\}^n$, the cipher $E: \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}^n$ is defined as:

$$E_k[P](x) = P(x \oplus k) \oplus k$$

- ▶ Minimal Construction
- ▶ Key ingredient in many symmetric-key constructions, e.g. Elephant, Chaskey



Example: Even-Mansour

- Block cipher from public permutation P : $E_k(x) = P(x \oplus k) \oplus k$
- SPRP Security in IPM via H-coefficient:
- B = set of transcripts where an input (output) of E_k/R differs from an input (output) of P by k
- Transcript probability very similar (more values excluded for real world)

The H-coefficient technique

The Swiss army knife of provable symmetric-key cryptography

QIPM changes everything

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QIPM changes everything

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Superposition oracles can help, but...

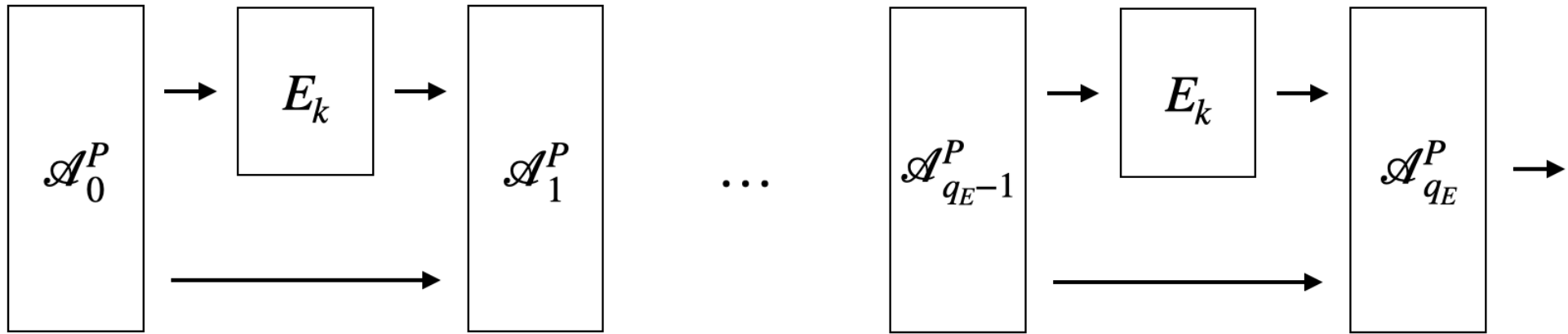
- Zhandry's compressed oracle technique: kind of a transcript
- Permutations: more difficult (**M**, Malavolta, Walter, STOC '25; Alagic, Carolan, **M**, Tokat, FOCS '25; Carolan, STOC '26)

Superposition oracles can help, but...

- Zhandry's compressed oracle technique: kind of a transcript
- Permutations: more difficult (**M**, Malavolta, Walter, STOC '25; Alagic, Carolan, **M**, Tokat, FOCS '25; Carolan, STOC '26)
- Crucially, still out of reach:
 - T determines the output bit
 - RVs T_0, T_1
 - $\Pr[T_i = \tau] \in \{0, p_i(\tau)\}$ where p_i only depends on $\mathcal{O}_i$
- Natural analogue of $\frac{\Pr[T_0=\tau]}{\Pr[T_1.=\tau]}$: Löwner order ⚡ superposition oracle

Alternative: Hybrids!

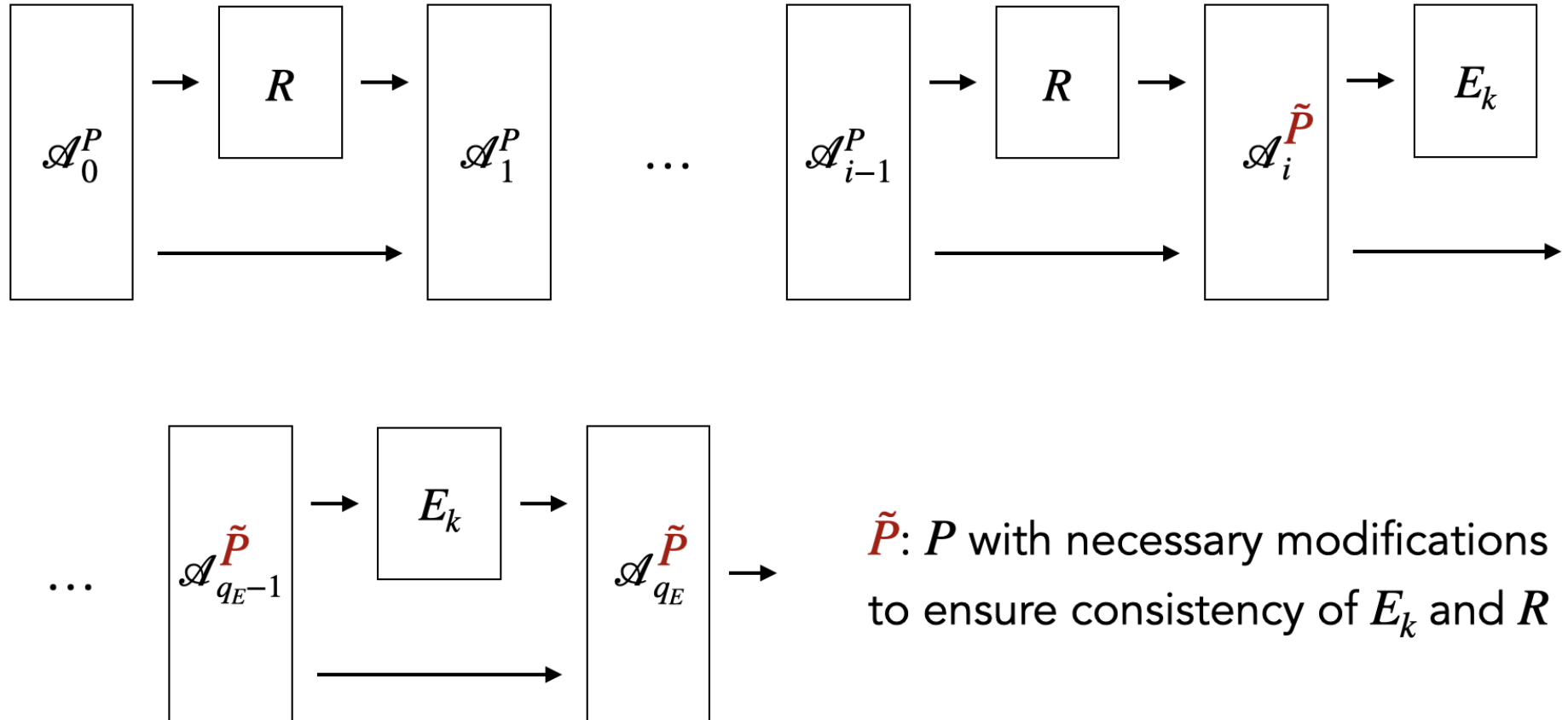
- ▶ Think of adversary having $q_E + 1$ stages:



- ▶ Each $\mathcal{A}_i^P$ makes arbitrary number of quantum queries to P , sum $\leq q_P$
- ▶ Naive hybrids: replace E_k with R for first i calls
- ▶ Consistency of the oracles?

Alternative: Hybrids!

H_i :



So $\tilde{P}$ gets messed up more and more as i grows... but in the end it's gone!

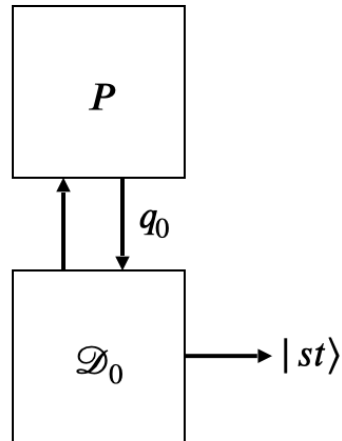
Indistinguishability of hybrids

- Two steps in hybrid transition:
 - Replace E_k with R and “repair” $\tilde{P}$
- Replace the first $\tilde{P}$ with P
- 2 Technical lemmas:
 - A **resampling lemma** for random permutations
 - A **reprogramming lemma** in terms of expected number of queries

Resampling

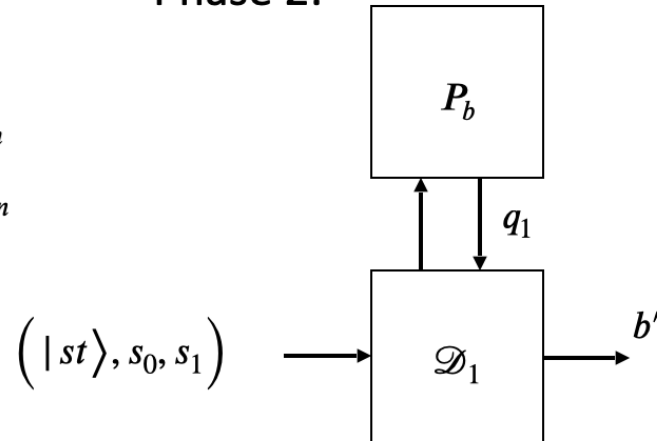
Different applications (random oracles, permutations, ideal ciphers) and different assumptions (uniform/high entropy/adaptive choice of input)

Phase 1:



Phase 2:

$$\begin{aligned} b &\stackrel{\$}{\leftarrow} \{0,1\} \\ s_0 &\stackrel{\$}{\leftarrow} \{0,1\}^n \\ s_1 &\stackrel{\$}{\leftarrow} \{0,1\}^n \end{aligned}$$



$\mathcal{D}$ wins if $b = b'$

$$P_0 = P$$

$$P_1 = P \circ \text{swap}_{s_0, s_1} = \begin{cases} P(s_0) & \text{if } x = s_1 \\ P(s_1) & \text{if } x = s_0 \\ P(x) & \text{otherwise} \end{cases}$$

$$\text{Lemma: Advantage} \leq O\left(\sqrt{q \cdot 2^{-n}}\right)$$

Reprogramming

Reprogramming game:

1. Distinguisher $\mathcal{D}$ supplies function F and specifies randomized reprogramming routine
2. For random bit b , $\mathcal{D}$ receives oracle F_b ($F_0 = F$ and F_1 is reprogrammed)
3. $\mathcal{D}$ loses access to F_b , receives description of F_1 , outputs b'

$\mathcal{D}$ wins if $b = b'$

min-entropy
of reprogramming

Lemma: Advantage $\leq O\left(q\sqrt{\varepsilon}\right)$

Refined Hybrids

$\mathbf{H}_0$: $|P\rangle, E_k[P], \dots$ $E_K[P], |P\rangle,$ $\dots, E_K[P], |P\rangle$

$\vdots$

$\mathbf{H}_j$: $|P\rangle, R, |P\rangle, \dots, R, |P\rangle,$ $E_K[P^j], |P^j\rangle,$ $E_K[P^j], |P^j\rangle, \dots, E_K[P^j], |P^j\rangle$

$\mathbf{H}_j^1$: $|P\rangle, R, |P\rangle, \dots, R, |P\rangle,$ $E_K[P^{j+1}], |P^{j+1}\rangle,$ $E_K[P^{j+1}], |P^{j+1}\rangle, \dots, E_K[P^{j+1}], |P^{j+1}\rangle$

$\mathbf{H}_j^2$: $|P\rangle, R, |P\rangle, \dots, R, |P\rangle,$ $R, |P^{j+1}\rangle,$ $E_K[P^{j+1}], |P^{j+1}\rangle, \dots, E_K[P^{j+1}], |P^{j+1}\rangle$

$\mathbf{H}_{j+1}$: $|P\rangle, R, |P\rangle, \dots, R, |P\rangle,$ $R, |P\rangle,$ $E_K[P^{j+1}], |P^{j+1}\rangle, \dots, E_K[P^{j+1}], |P^{j+1}\rangle$

$\vdots$

$\mathbf{H}_{q_c}$: $|P\rangle, R, |P\rangle \dots$ $R, |P\rangle,$ $\dots, R, |P\rangle$

$\underbrace{\hspace{10em}}$

$\underbrace{\hspace{10em}}$

$\underbrace{\hspace{20em}}$

j classical queries

$(j + 1)^{\text{st}}$ classical query

$q_c - j - 1$ classical queries

Result:

Theorem (Alagic, Bai, Katz, M):

Let P and R be a random permutations and E_k the Even-Mansour cipher using P . Any quantum adversary making at most q_P quantum queries to P and q_E classical queries to an oracle O has distinguishing advantage between $O = E_k$ and $O = R$ at most

$$10 \cdot 2^{-\frac{n}{2}} \left(q_P \sqrt{q_E} + q_E \sqrt{q_P} \right)$$

- First term is tight, second is not
- Matching attack for first term: BHT, Offline Simon's

Applications

- Post-quantum Security of the **Even-Mansour cipher** [[ABKM22](#)]
- Post-quantum Security of **Tweakable Even-Mansour**, and Applications [[ABKMS24](#)]
- Parallel-query post-quantum Security of **Key-Alternating Ciphers** [[BEM25](#)]
- Post-quantum Security of **Key-Alternating Feistel Ciphers** [[BBCJJRST25](#)]
- Pseudorandom Function-Like States from Common Haar Unitary [[HY25](#)]
- Post-quantum security of **keyed sponge-based constructions** [[Hos25](#)]
- Post-quantum Security of **FX, LRW, XEX** [[ABMS26](#)]
- **Multi-Key Security** in the Quantum World: Revisiting Tweakable Even-Mansour and FX [[SI26a](#)]
- Post-quantum Security of **Tweakable Key-Alternating Feistel Ciphers** in the Multi-key Setting [[SI26b](#)]
- Post-quantum multi-key security of **GCM** (in QICM) [[Hos26](#)]
- More ...?

Take-home messages

- Post-quantum security of symmetric-key crypto should be fine, but work is still needed to provide the same level of provable security guarantees as classical
- H-coefficient seems unlikely to be of help in the post-quantum setting
- The potential of the ABKM22 hybrid technique doesn't seem exhausted yet



Thank you